Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/04/2021
PI Name:	Sekulic, Dusan D.Sc.		
Project Title:	BRAzing of Aluminum Alloys IN Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	40506	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/04/2016	**End Date:**	09/30/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Hanson, Ian	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 09/30/2022 per NSSC information (Ed., 11/27/21) NOTE: End date changed to 11/3/2021 per NSSC information (Ed., 3/30/21)		
Key Personnel Changes/Previous PI:	September 2021: No changes		
COI Name (Institution):	Krivilev, Mikhail Ph.D. (Udmurt State University, Russia) Mesarovic, Sinisa Ph.D. (Washington State University)		
Grant/Contract No.:	NNX17AB52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We have been investigating capillary flow phenomena relevant for techniques of brazing (bonding metallic or non-metallic mating surfaces by molten alloys) in microgravity for space construction and repair. A series of microgravity (International Space Station - ISS) and terrestrial experiments have been integrated with modeling and numerical simulations. These are proposed to assist in answering the practical and fundamental questions: (1) How can we take advantage of the absence of gravity and dominance of capillary forces in transporting the large quantity of molten braze? (2) What type of brazing alloy and brazing technique would enable strong and efficient bonding of structural elements and repair? (3) Can this brazing be done using flux-less cladding? The phenomena to be studied under these conditions include molten metal kinetics and wetting and spreading of liquid clad/filler over topographically modified surfaces. The applied aspects include direct verification if brazing of Al alloys is transferable to orbit for rapid repair of the pressure wall or construction. For the first time, brazing in space will be processed using the patented flux-less Trillium technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact of the proposed research on the life on Earth manifests itself through multifaceted benefits. These include: (i) uncovering so-far hidden physical/chemical/metallurgical aspects of metal bonding by brazing (extensively used in manufacturing) through uncoupling of the gravity impact on larger mass of molten aluminum driven by surface tension in gravity field, hence transforming the art of this technology to a predictive science for terrestrial applications, (ii) exploring a possibility to use novel materials as bonding alloys less sensitive on terrestrial hindrances (e.g., oxidation) in designing bonded assemblies and mitigating problems involving repair, (iii) featuring sustainable resources use and less energy invested due to an expected broadening the range of applicability of bonding/repair under adverse conditions, (iv) developing visualization of the brazing process in situ and in real time under controlled atmosphere, (v) developing the kinetics of wetting theory by implementing phase field approach and numerical techniques that can be used to model capillary phenomena not only for liquid metals but for other material systems in both microgravity and gravity (on Earth) conditions.
Task Progress:	This document constitutes the brief report of the research work conducted within the scope of the NASA sponsored Brazing Aluminum in Space (BRAINS) project (Grant #NNX17AB52G). The work has included both experimental and theoretical modeling. The participation of three collaborating teams, the University of Kentucky (UK), the Washington State University (WSU), and the Udmurt State University (UdSU), Russia, have been incorporated. The project was included into the long-term program of Roscosmos in November 2020 to support in-orbit implementation of microgravity research activity onboard of the ISS. Both experimental and theoretical (modeling) works have been executed as planned at UK, WSU, and UdSU. A surface tension driven flow of molten aluminum has been considered. Modeling includes kinetics of molten alloy filling the hole configurations and flow in a wetting-non-wetting systems. The ground and flight tests are ongoing. Due to the SpaceX flight schedules changes, the experimental runs at the ISS have been rescheduled. All data will be presented in technical papers published or to be published. We have discovered that the dynamics of the triple line (braze-gas-solid) on the non-wetting ceramics is controlled by physico-chemical ageing processes which alter the equilibrium contact angle. The modeling activities during the last year have been focused on the following tasks/issues: (1) Modeling of surface tension-controlled equilibrium formations and kinetics of the molten aluminum flow, (2) Modeling of the kinetics of the process of filling the holes by surface micro layers of liquid aluminum alloy and wetting-non wetting substrates systems. Theoretical (modeling) work has followed a successful development of the phase field theory of the dynamic behavior of liquid metal during a hole-filling and-or a corner flow of a wetting-non-wetting system flow. Results from phase field models agree very well with empirical data.
Bibliography Type:	Description: (Last Updated: 12/06/2025)
Abstracts for Journals and Proceedings	Lazaridis K, Wu Y, Mesarovic SDj, Sekulic DP. "Benchmarks for capillary flows under terrestrial and zero gravity conditions." 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Wu Y, Lazaridis K, Sekulic DP, Krivilyov MD, Mesarovic SDj. "Kinetics of the triple line on wetting and non-wetting surfaces: relaxation and ageing." 14th World Congress in Computational Mechanics (WCCM) ECCOMAS Congress 2020, Paris, France, 1-15 Jan 2021. Abstracts. 14th World Congress in Computational Mechanics (WCCM) ECCOMAS Congress 2020, Paris, France, 1-15 Jan 2021. , Jan-2021
Abstracts for Journals and Proceedings	Lazaridis K, Wu Y, Krishna Muniyal SK, Yu C-N, Krivilyov MD, Sekulic DP, Mesarovic SDj. "Ageing of equilibrium contact angle in capillary flows." 5th Serbian-Greek Symposium on Advanced Mechanics, Kragujevac, Serbia, June 28-30, 2021. Abstracts. 5th Serbian-Greek Symposium on Advanced Mechanics, Kragujevac, Serbia, June 28-30, 2021. , Jun-2021
Abstracts for Journals and Proceedings	Lazaridis K, Wu Y, Krishna Muniyal SK, Yu C-N, Krivilyov MD, Sekulic DP, Mesarovic SDj. "Anomalous capillary flow of a molten metal – Ageing contact angle." 74th Annual Meeting of the APS Division of Fluid Dynamics, Phoenix, Arizona, November 21-23, 2021. Bulletin of the American Physical Society. 2021, in press as of October 2021. , Oct-2021
Articles in Peer-reviewed Journals	Voroshilov E, Krivilyov M. "Simulation of wetting of a solid of revolution by viscous liquid." AIP Conference Proceedings. 2020;2216(1):030009. Published Online: 01 April 2020. https://doi.org/10.1063/5.0003586 , Apr-2020
Articles in Peer-reviewed Journals	Shutov IV, Kamaeva LV, Khamidullina AR, Krivilyov MD, Sekulic DP. "Phase transformations under heat treatment of Al-Si + flux composite brazing metal." Metal Science and Heat Treatment. 2020 Nov; 62(7-8):498-501. https://doi.org/10.1007/s11041-020-00591-4 , Nov-2020
Articles in Peer-reviewed Journals	Wu Y, Yu C-N, Sekulic DP. "Si diffusion across the liquid/solid interface of capillary driven (Al–Si)-KxAlyFz micro-layers." Journal of Materials Science. 2021 Apr;56(12):7681-97. Published: 03 January 2021. https://doi.org/10.1007/s10853-020-05689-x , Apr-2021

Articles in Peer-reviewed Journals	Yu C-N, Lazaridis K, Wu Y, Voroshilov D, Krivilyov MD, Mesarovic SDj, Sekulic DP. "Filling a hole by capillary flow of liquid metal–equilibria and instabilities." Physics of Fluids. 2021 Mar;33(3):034109. https://doi.org/10.1063/5.0039718 , Mar-2021